



WINGSPAN

An options agent that mostly says no.

DeepSeek proposes. Deterministic Python decides every strike, size and exit.
The official Alpaca CLI places every order. Every refusal is journaled.

Alpaca AI Trading Agents Hackathon 2026 · team Convexity · paper account PA371G5THNUO

The problem

- Language models are good at generating plausible trades and bad at refusing bad ones.
- In a live account the expensive mistakes are not the ideas, they are the sizes, the strikes and the exits.
- So the model only answers one question per name: is there a trade here, which strategy, which direction, how sure?
- Everything after that is deterministic code that can be tested, replayed and audited.

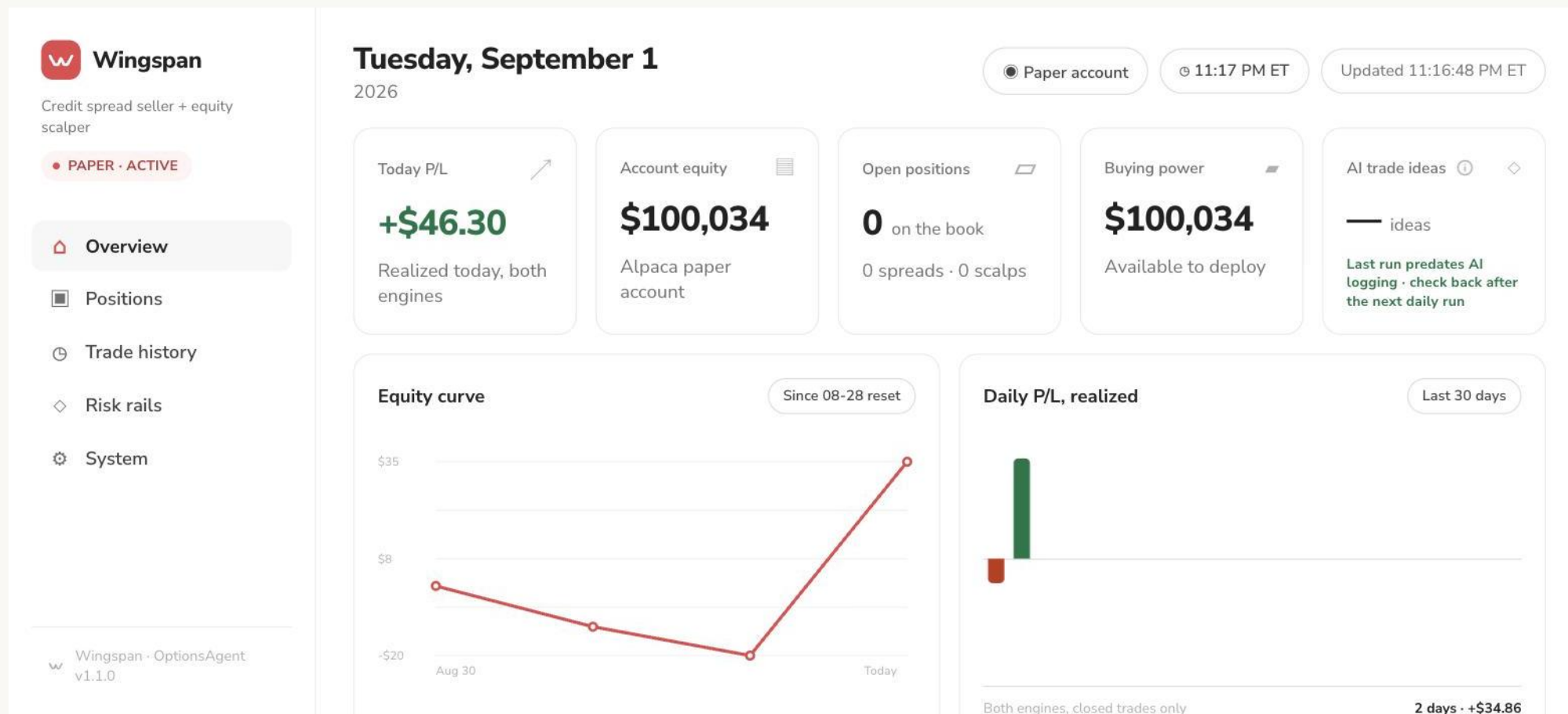
The rule: AI proposes, code disposes

- Once a day DeepSeek reads a 13-name watchlist with fresh bars and context, returns {underlying, strategy, direction, conviction, thesis}, validated against a schema.
- Deterministic rails: conviction floor 0.60, strategy menu, one position per name, six slots, a \$3,000 hard cap per position.
- Contract picker: short strike 0.15 to 0.30 delta, 30 to 45 days out, width at most \$2. Liquidity floor: credit $\geq$ \$0.10 and unwind cost $< 1.5x$ credit.
- One multi-leg limit order through the official Alpaca CLI. Fill confirmed, remainder cancelled, filled count booked.

Exits have no AI in them

- Profit target: buy the spread back at 50% of the credit received.
- Stop: cost to close at 2x the credit, only after 10:00 ET and only when two consecutive sweeps agree, so one wide opening quote cannot force a liquidation.
- Time: forced close at 21 days to expiry, before gamma risk grows.
- A close is recorded only when the unwind order actually fills; a partial fill re-registers the remainder.

Live dashboard



Risk rails, as the code sees them

 **Wingspan**

Credit spread seller + equity scalper

PAPER · ACTIVE

Overview

Positions

Trade history

Risk rails

System

Wingspan · OptionsAgent v1.1.0

Tuesday, September 1 2026

Paper account

11:17 PM ET

Updated 11:16:48 PM ET

Risk rails

Every gate below is hard-coded in Python. The AI only proposes; these rails decide, and they can be tightened by environment but never loosened.

← Overview

Credit spread seller

Engine

Runs

AI proposer

Strategy phase

Minimum conviction

Max open option legs

Max per position

Enabled

Once a day, 10:15 ET, weekdays

deepseek · deepseek-v4-pro

Credit Spreads Only

60%

6

\$3,000

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Alpaca infrastructure

- Account, positions, clock, order submit (single-leg and mleg), order status and cancel all run as `alpaca ... -q` subprocesses inside the container, JSON out.
- CLI pinned to v0.0.14 with a verified checksum in the Docker image. Every call journaled: argv, exit code, latency, order id.
- Fail-closed: a missing binary or a non-zero exit stops the trade. A lost reply is reconciled by our own client order id before it is called a failure.
- Runs on Railway as Linux cron: entry 10:15 ET, exits every 20 min, an equity scalper every minute. State on a volume. Paper only by construction.

What we caught before the first trade

- The original gate admitted only three historical winner shapes: a week of zero option fills.
- Opening it naively would have booked losers: replaying the previous day's real chains, 8 of 22 admitted spreads were already past the 2x stop on their own entry quotes, because entry is priced bid-to-ask and the unwind ask-to-bid.
- Fix: a liquidity floor derived from the exit rule. After it: 5 of 26 admitted, 0 past the stop.
- Also fixed before deploy: no per-name dedupe, fills booked unconfirmed, a scalper path that bypassed the CLI. 276 tests; seven mutation checks each broke the right test.

Results and disclosure

- Competition account PA371G5THNUO: equity \$100,034.43; today's P/L \$46.30.
- Filled orders in the window: 6 (0 options, 6 shares).
- Active settings: gate = research_rules, cap = \$3,000.00, broker path = cli, proposer = deepseek · deepseek-v4-pro.
- Disclosure: the harness predates the event (July 2026). Built in the window: the equity scalper and its study, the Railway deploy, the DeepSeek proposer, the dashboard rework, the CLI transport, the gate and cap. Nothing else has traded this account.
- Repo: github.com/Jhosshua/OptionsAgent (MIT). Dashboard: optionsagent-production.up.railway.app